

ITAI 1371 – Module 05

Contribution Journals

Combined by student name:

- Alexander DeBusk
- Shareefah Adeola
- Yilin Leng

Lab 5 Contribution

- Reviewed the group leader's instructions and completed my assigned portion of the lab.
- Completed the required data preparation tasks, including missing-value imputation, categorical encoding, and feature scaling.
- Submitted my completed work to the group leader on time.

Contribution Journal – Module 05: Data Preparation

ITAI 1371

Student: Shareefah Adeola

Date: 09/20/2026

An important part of my contribution was **troubleshooting and problem-solving**. I encountered a syntax error at the beginning of the notebook because some Python statements had been combined incorrectly. I corrected the formatting and reran the code successfully. I also encountered a NameError indicating that pd was not defined. I determined that the setup cell needed to be executed again so that the pandas library and dataframe were available. After rerunning the necessary cells, I continued with the assignment successfully.

I also reviewed the output after each major step rather than assuming that the code had worked.

For example, I verified that the missing Age values changed from 177 to zero, checked the new encoded columns after One-Hot Encoding, and reviewed the standardized Age and Fare values after Feature Scaling.

Finally, I completed the Knowledge Check questions in my own words. I explained why the median can be preferable to the mean when data contains outliers, how One-Hot Encoding converts categorical information into numerical features, and why Feature Scaling is generally not required for a Decision Tree.

My Overall Contribution

My contribution to this module involved **coding, data preparation, testing, troubleshooting, and documenting my understanding of the workflow**. Completing the tasks independently helped me better understand how missing-value treatment, categorical encoding, and feature scaling work together.

This experience also showed me that contributing to a machine learning project is not only about writing code. Checking results, identifying errors, understanding why a technique is being used, and documenting the work are also important parts of the process. I completed my assigned work and verified that the notebook could run successfully from beginning to end without errors.

Contribution

During this project, I worked independently on several important parts of the data preparation process. My main contributions included handling missing values, encoding categorical data, scaling numerical features, and completing the Knowledge Check questions.

I worked on the missing values in the "Age" column of the Titanic dataset. There were 177 missing values, so I used the median to fill them in. I also used `.isnull().sum()` to check the data before and after the process. After the imputation, I confirmed that there were no missing values left in the "Age" column.

Working on converting categorical features into numerical data. I used `pandas.get_dummies()` to encode the "Sex" and "Embarked" columns. I used `drop_first=True` to remove one of the repeated categories and help avoid multicollinearity. This made the categorical data easier for machine learning models to use.

Another part I completed was scaling the numerical features. The "Age" and "Fare" columns had very different ranges, so I used `StandardScaler` from `scikit-learn` to standardize them. This changed the features to a similar scale, with a mean of 0 and a standard deviation of 1.

And I also completed the Knowledge Check questions based on the work from the experiment. I explained why the median can be useful for missing values, why One Hot Encoding is needed for categorical data, and why decision trees do not require feature scaling. Completing these questions helped me connect the coding steps with the concepts behind them.

Overall, my main contribution to this project was working on the data preprocessing steps and checking that the results were correct. I also used the results from the experiment to better understand why each preprocessing method was used.